

Session 2: Competitive Advantage Tesla

Competitive Strategy
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How is our 15% class presentation graded? (22 Jan 2025)

1. Quality of presentation (least important)
2. Analytical content.
3. Tools and framework from course.

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Within-Industry Performance Differences

Fraction of variance of plant-level growth rates explained by detailed industry, U.S. mfg 1977-87 (industry examples: chewing gum, luggage mfg., saw blade mfg.; 450 in mfg.)

Outcome—Growth Rate of:	Fraction of Variance Accounted for by Industry
Employment	
Capital equipment	
Capital structures	
Revenue	
Labor productivity (output/hr)	
Total-factor productivity	

Within-Industry Performance Differences

Fraction of variance of plant-level growth rates explained by detailed industry, U.S. mfg 1977-87 (industry examples: chewing gum, luggage mfg., saw blade mfg.; 450 in mfg.)

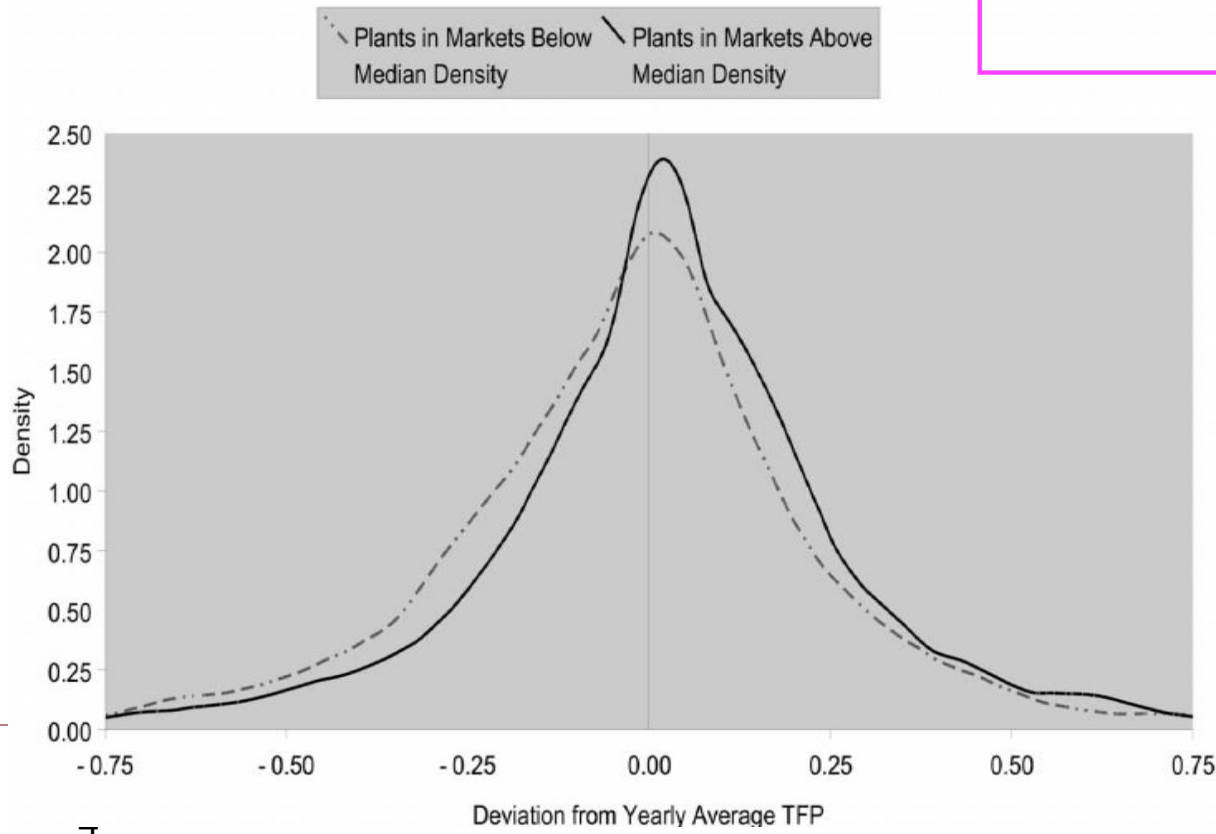
We see big & Persistent Performance Differences between firms.
Why?

Outcome—Growth Rate of:	Fraction of Variance Accounted for by Industry
Employment	5.7%
Capital equipment	6.2%
Capital structures	5.2%
Revenue	8.9%
Labor productivity (output/hr)	8.6%
Total-factor productivity	9.5%

Within-Industry Productivity Dispersion

TFP distribution, U.S. ready-mixed concrete industry (about 15,000 plant observations)

Total Factor Productivity.
- dense market, distribution is sharper. (more competitive)
-



Within-Industry Performance Differences

Summary measures of within-detailed-industry TFP (output per unit input) variation, U.S. mfg

~450 industries

TFP comparison	Average Within-Industry TFP Ratio
75 th %ile vs. 25 th %ile	1.34:1
90 th %ile vs. 10 th %ile	1.92:1
95 th %ile vs. 5 th %ile	2.55:1

Summary

Performance differences across firms are:

1. Huge, even narrowly defined industries
2. Persistent (luck will not save you)
3. Predictors of a firm's fate. Tied to higher:
 1. Profitability
 2. Growth rates
 3. Survival probability (i.e., a matter of life and death)

Patterns found in every industry, country, and time period

Individual firm's strategy matters alot.

What is Competitive advantage?

In strategy a source of performance differences is called a “competitive advantage” (CA)

(scope) On a particular segment, for a particular set of uses

(comp. adv) We provide higher wtp/lower cost than our rivals

Types of Competitive advantage

Demand-side

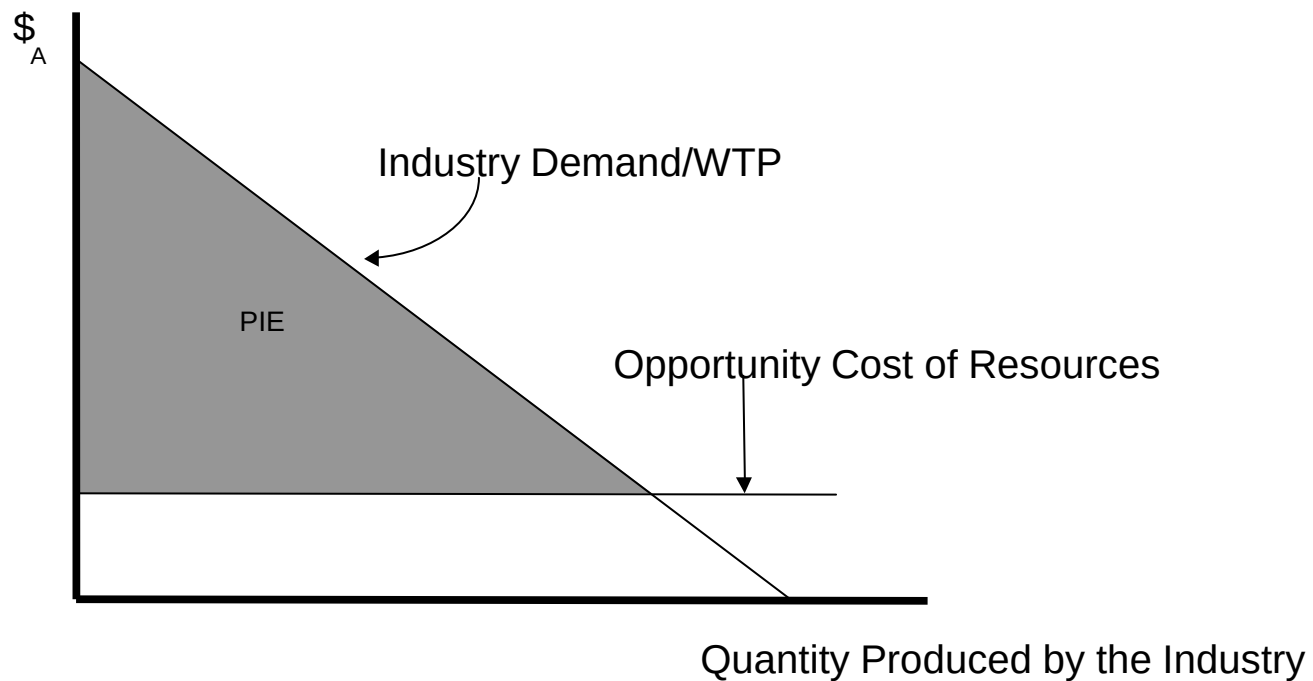
- Does it affect (some) consumers' willingness to pay (WTP) for your product relative to competitors' products?

Cost-side.

- Does it enable you produce a product at a lower cost than competitors can or could?

A combination of both is particularly powerful

Identifying PIE



Value Division

Customer

Willingness to pay

Value captured by customer

Firm

Price

Value captured by firm

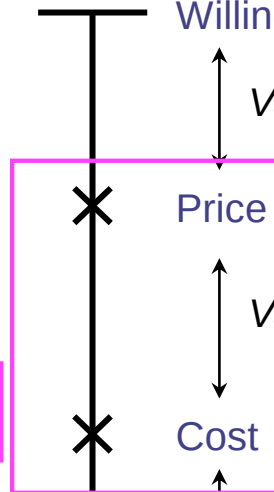
Cost

Value captured by supplier

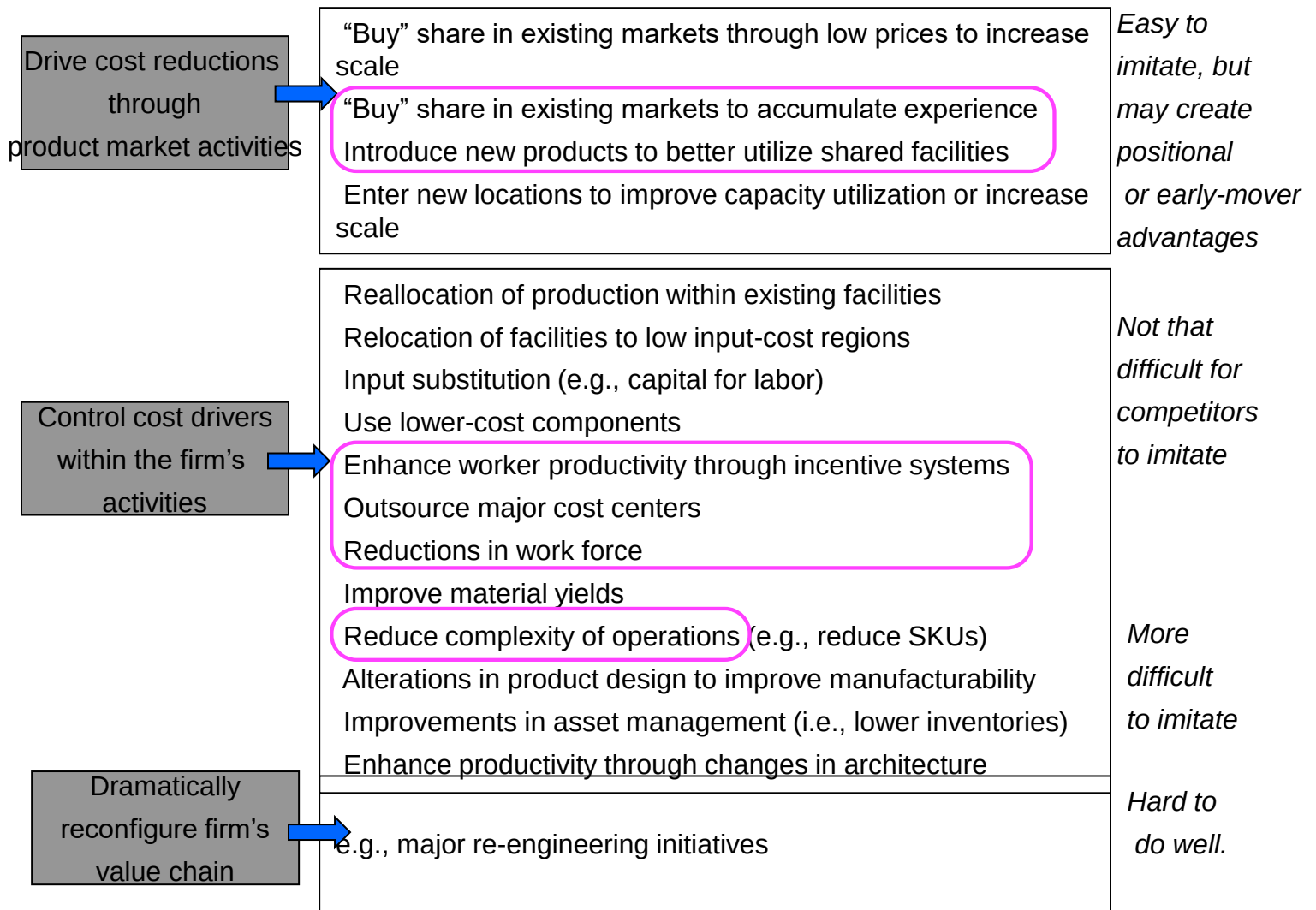
Supplier

Supplier opportunity cost
(willingness to sell)

How to capture the differences here?



Lower Cost: Drivers



Higher WTP: Drivers

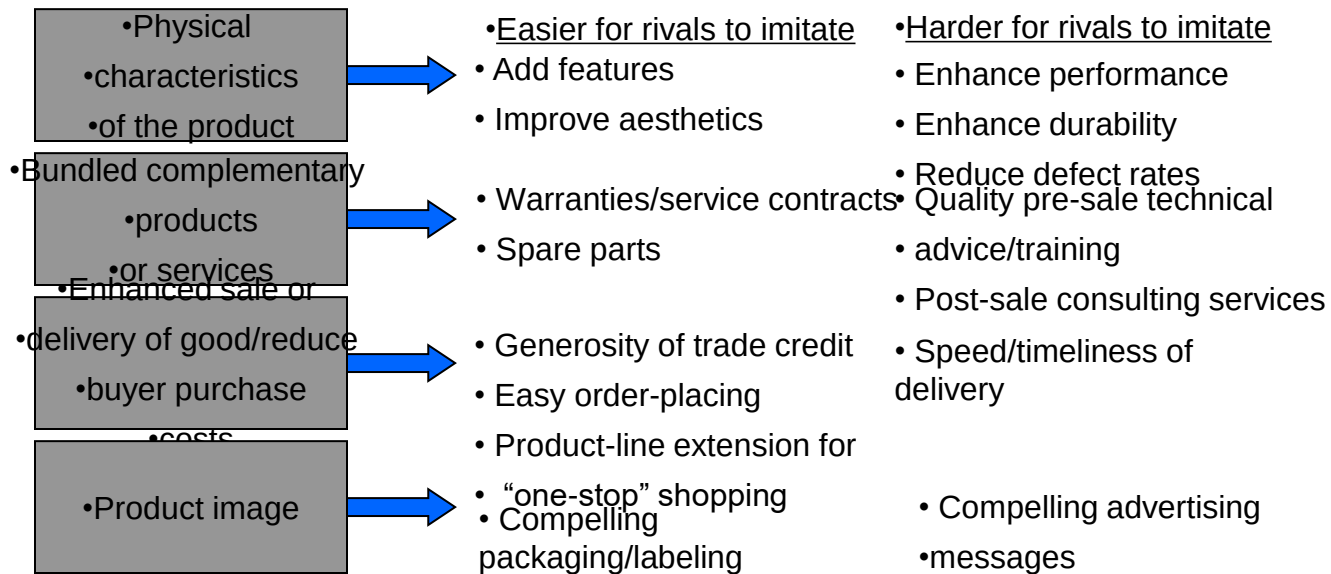
Product as a bundle of attributes

- **Physical characteristics**
 - e.g. quality, durability, aesthetics
- **Quantity and characteristics of complementary goods and services**
 - e.g. after-sale, consulting, spare parts,...
- **Attributes associated with sale and delivery**
 - e.g. credit, delivery, location of seller,...
- **Characteristics that shape consumer's perception expectations**
 - (e.g. reputation, financial stability, installed base)
- **Subjective image of the product**

Identify what are the key WTP drivers in my industry?

- specialisation
- quality of candidate
- relationship
- bundled services
- MOUs with key partners

Drivers of WTP advantage



3 Sources of Competitive Advantages

Position-based Competitive Advantage

Football, i can score a goal just by being in the right position (even though i am not good)

Capability-based Competitive Advantage

My capabilities as a footballer is superior.

Resource-based Competitive Advantage

i have better boots and equipments to grant me favorable advantage.

Position-based Advantages: Logic

Dominate a market or market-segment

Competitive advantage relies on a combination of
Economics of Scale:

- Rivals/entrants need to achieve a substantial market share in your segment – otherwise they will be at a competitive disadvantage relative to you!

Customer Captivity

- Due to Switching costs, Search Costs, Habit formation, Brand loyalty, it is difficult for rivals to quickly build market share
- As long as you match competitive moves (prices, product features) by rivals, customers unlikely to defect from you!

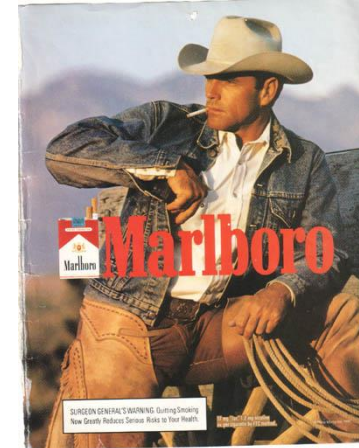
Positional advantages



We'll pick you up.®



*Do I have, or can I develop,
a superior position?*



Position-based Advantages: Logic

Position-based advantages often associated with “early-mover” advantages

Position-based Advantages often stem from dominating a “niche” or “market segment”



Positional Advantage

Ideally need two conditions

- Some stickiness on the demand side:
 - Consumer switching costs
 - Network effects
- Economies of Scale large relative to market size

Why both?

- Without stickiness, entry possible by achieving scale, and then fight on equal conditions
- Without Economies of Scale, can start small and develop market little by little

On the Cost Side: Economies of Scale, Sunk costs

As in Industry Analysis, but instead of protecting an industry they protect a segment or position..

Imitator tries to compete on your terms (low unit costs via large sales volumes in industry with large MES relative to demand).

Imitator must exceed the minimum efficient scale. But how?

- He can't drive you out of business. You will stay in as long as you can cover your variable costs!
- Because MES is large relative to market demand, prices must fall significantly in order to absorb production of both
- The NPV on the investments he needs to make to copy the basis of your cost advantage are likely to be negative. If he's smart, he'll stay out!

Market demand can only support 1 big store, but not 2. So the entrant faces very light MES, and must suffer long duration of losses.

And on the Demand Side: Switching costs

Switching Costs: stem from the loss of capital (physical or human) that was acquired through use of a particular product or service

Investments buyers make in sellers product:

- e.g. machine that fits with seller's product

Supplier specific learning:

- e.g. training in Windows

Contractual switching costs:

- e.g. frequent flier programs

Uncertainty: brand loyalty / quality reputation:

- e.g. Disney animated movie and kids

Psychology: order of entry matters:

- People trust the innovator: e.g. Apple and iPod

Installed Base Advantage with Network Externalities

Network externalities exist when a product's "B" to a given consumer increases when more consumers use the product or are expected to adopt it in the future.

Why?:

- Direct: products fit together
- Indirect: Complementary goods supply

Indirect NW Effect or NW Externality. Example; Whatsapp. I will use it because more of my contacts are on it. Same for FB, IG, Mac,.

Liquidity of platform - WTP buyers increases number of buyers, then increase WTP sellers which increase number of sellers.

Direct NW; WTP increases $\rightarrow$ # and Q increase $\rightarrow$ WTP increase further.

Because you got into the market early, your business unit has a larger installed base than other competitors.

- The smaller installed bases of latecomers makes their products less attractive than yours at similar prices. (Windows phone)
- Plus own large installed base makes it more attractive for suppliers of complementary products (apps in iPhone)
- This adds to the installed base, making the product even more attractive to consumers in the future.

3 Sources of Competitive Advantages

Position-based Competitive Advantage

Capability-based Competitive Advantage

Resource-based Competitive Advantage

Capabilities as a source of competitive advantage

Learning Curve Advantages

Organizational capabilities



Learning Curve

Because you got into the market early, business unit has accumulated important experience and knowledge for production

Learning curve: Costs fall with cumulative output due to increases in efficiency

- if too steep, followers catch up quickly
- if too flat, cost advantages from learning not large

BUT: inter-firm technology diffusion limits learning advantages

- inter-firm mobility of workers , technological communication, reverse engineering etc.

Learning curve steepness for services industry. Like consultancy? Hairdresser?

https://en.wikipedia.org/wiki/Experience_curve_effects

Experience that Prof keep saying; DeepSeek and LLMs.

– many service sectors (including AI business) have very flat learning curve.

DeepSeek spent 6M to achieve certain competencies.

<https://en.wikipedia.org/wiki/DeepSeek>

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If too steep, there is very little time needed for competitors to catch up.
75% to 95%. When you duplicate a production, how much cost goes down.

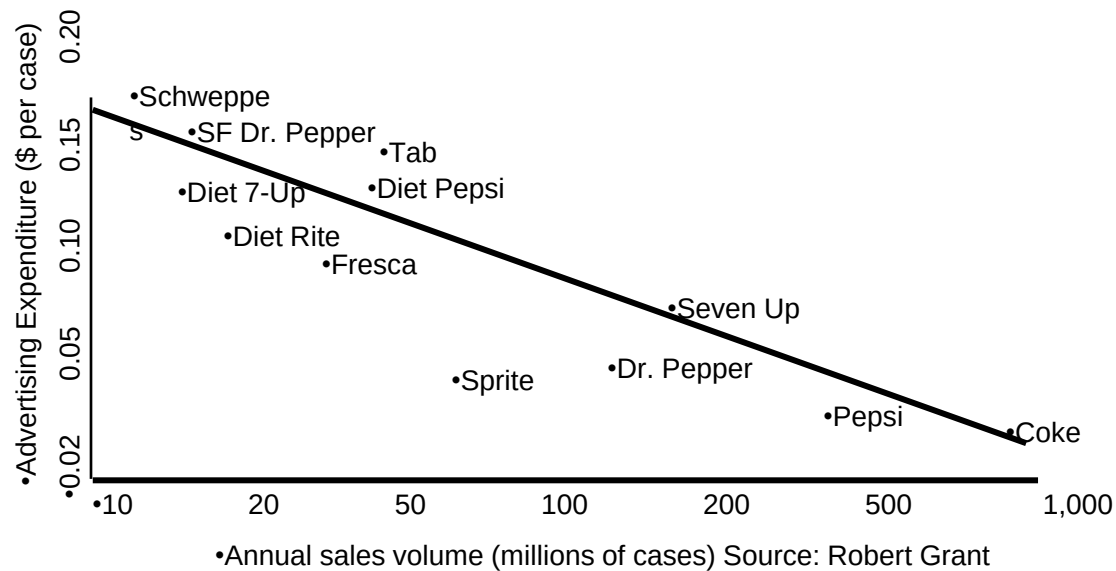
e.g, for 2nd plant, the cost is $75\%^2$. this is reasonable good.

Learning Curve vs. EOS

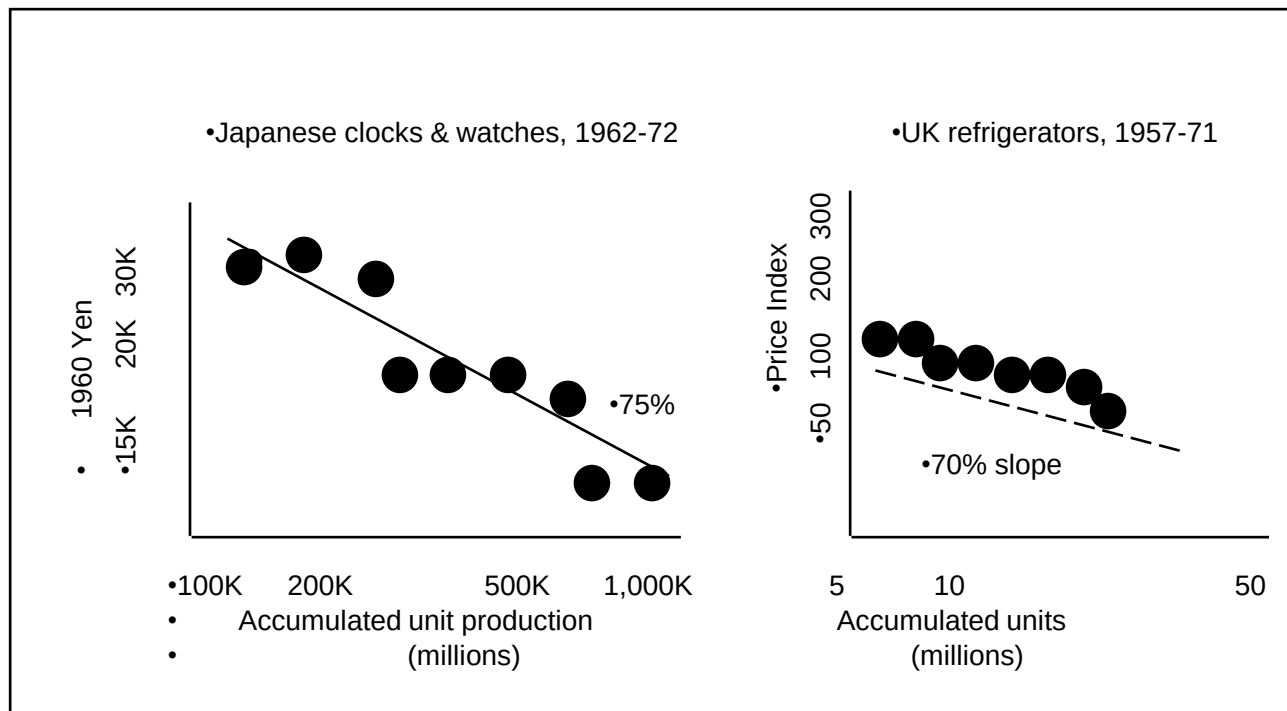
Learning (or experience) curves: Unit cost decreases with total number of units produced

EOS: Unit costs decrease with output per unit of time (e.g. cars per year)

Economies of Scale: Advertising, US soft drinks



Learning Curves: example



•Source: Robert Grant

Lesson: size within niche matters

When EOS or Learning curves important, market share is crucial

EOS:

- Lower unit costs until MES is reached

Learning curves

- With same years of experience- the firm with a higher share has more actual experience

3 Sources of Competitive Advantages

Position-based Competitive Advantage

Capability-based Competitive Advantage

- Learning curve/experience
- **Organizational design.**

Resource-based Competitive Advantage

Organizational Capabilities

Analogy to individual skill:

- Just as human beings differ in their skills (due to initial endowments and their histories), so do organizations.

Key difference: capability must be more than a simple summation of employees' skills.

- Skilled employees have bargaining power!
- So, to be the basis for an organizational capability, the same individuals must be more valuable in the firm than in another.

Much harder to define and learn Org Capabilities.

- skilled employees are bound to capture some (or most) of the value created.

-

Organization Capability -- ARC

- The organizational designer must think about all aspects
 - Architecture (formal and informal organization, incentive systems)
 - Routines (regularized ways of performing activities and coordinating),
 - Culture (shared understandings and beliefs)
- and should deal with these challenges in a manner that highly consistent, both internally and externally: complementarities

Key components of culture; tacit collabs, understandings, leadership styles..
- Best way to measure culture is; who gets promoted? do they only promote the collaborators, the curry-favors, the legacy folks?

Routines; general practices

IMPORTANT; how to identify what are easy or hard to change.

ARC Framework

Architecture

- Org chart, functional v. divisional structure, incentive schemes, & so forth

Routines

- Day-to-day activities, interfaces between parties for coordinating activities & allocating resources

Culture

- Commonly held values/beliefs that constrain & enable actions

ARC Framework

Architecture

- Org chart, functional v. divisional structure, incentive schemes, & so forth

Routines

- Day-to-day activities, internal control systems, coordinating activities & allocating resources

Culture

- Commonly held values/beliefs that constrain & enable actions

• **Easy to change**

• **Hard to change**

Understanding Organizational based Capabilities: complementarities

Two choice variables are **complements** if doing (more of) one, increases the returns to doing (more of) the other.

Example;

- promotions criteria -> employer branding / culture / retention, etc.

-

Two choice variables are **substitutes** if doing (more) of one, reduces the attractiveness of doing (more of) the other.

Complementarities: Example

Example:

Flexibility of
Manufacturing
System

Variety of Product
offerings
(breadth product
line,
frequency of
changes)

		FLEXIBILITY	
		Low	High
B R E A D T H	Low	•(Henry) •FORD	•GM
	High	•?	•Toyota

Why capability sustainable?

Key: complementarities.

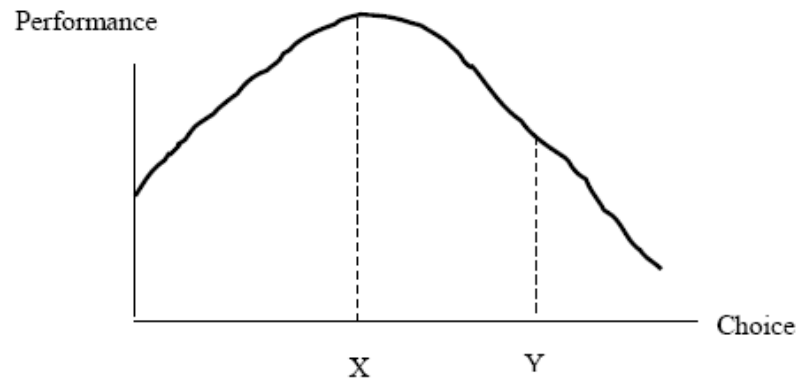
Lots of related, tightly linked choices

Generates profit landscapes with many peaks— hard to find and climb the tallest.

‘Causal ambiguity’ and ‘path dependence’

Non-concavities and Imitation

2 variables to toggle with to identify highest ROI on choice vs performance.



The classic Model of Choice: A Concave Performance Relationship

Finding the best choice is relatively simple, because local experiments that

- improve performance will eventually lead to the optimum.

Non-concavities and Imitation

There may be multiple choices (or patterns of choice, when there are many choice variables) that are coherent.

Yet among these multiple coherent patterns, some yield better performance than others do

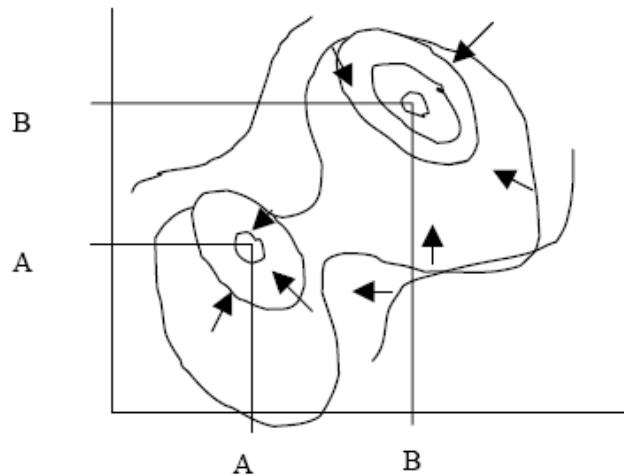
When choice is multi-dimensional, then no change, however big, in a subset of the variables can improve performance

No small adjustment in the set of choices can increase Performance (locally best choice)

Non-concavities and imitation

From peak A to peak B, you are going through a 'desert / valley'. Profits will decline first. But how do you marshal your management and staff to go through this?

- it may involve full scale restructuring, retrenchment, changing operational processes etc.



- Non-concavity allows multiple, distinct coherent points

Complementarities and imitation

“Causal ambiguity”

- Difficult for an outsider to appreciate how the elements of the organization’s ARC combine to produce the capability.

Architecture, Routine, Culture.

Casual ambiguity;

Two main reasons:

- Complexity: the system of complementarities is too complex to understand and copy.
- Tacit knowledge: system is not codified but embedded in largely unstated routines and shared, but implicit, understandings by insiders.
 - It may even be hard for insiders to really grasp what makes the system work (hence Intel’s approach to constructing new fabs.)

Complementarities and imitation

Experience clearly important too.

- Organizational capabilities can rarely be created from scratch but typically result from a long process of trial-and-error experimentation (experience curve).
 - One of the reasons for the “liability of newness” that plagues all start-ups.
- History has its down side too: can lead to “liability of obsolescence” whereby older organizations are “trapped” in areas of established competence.

3 Sources of Competitive Advantages

Position-based Competitive Advantage

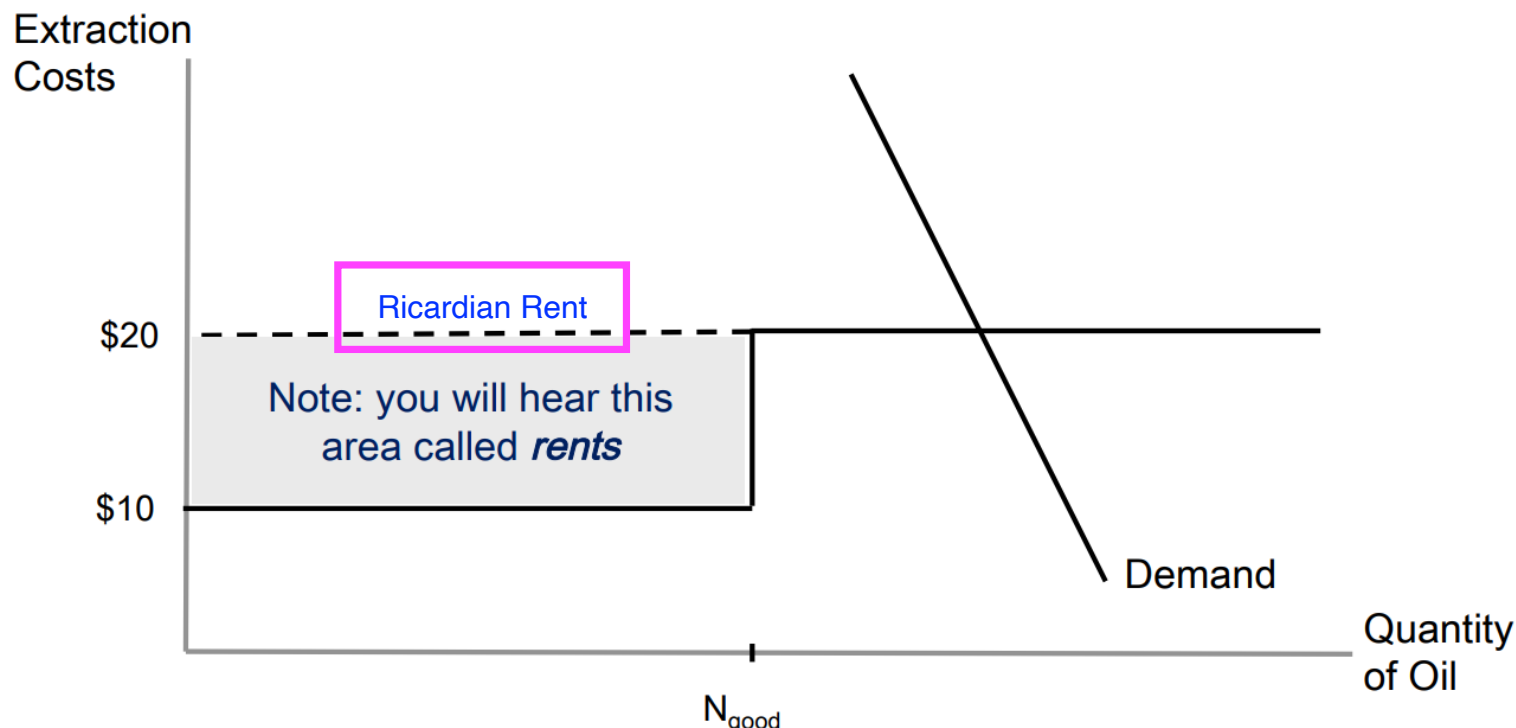
Capability-based Competitive Advantage

Resource-based Competitive Advantage

Scarce and Valuable assets earn rents

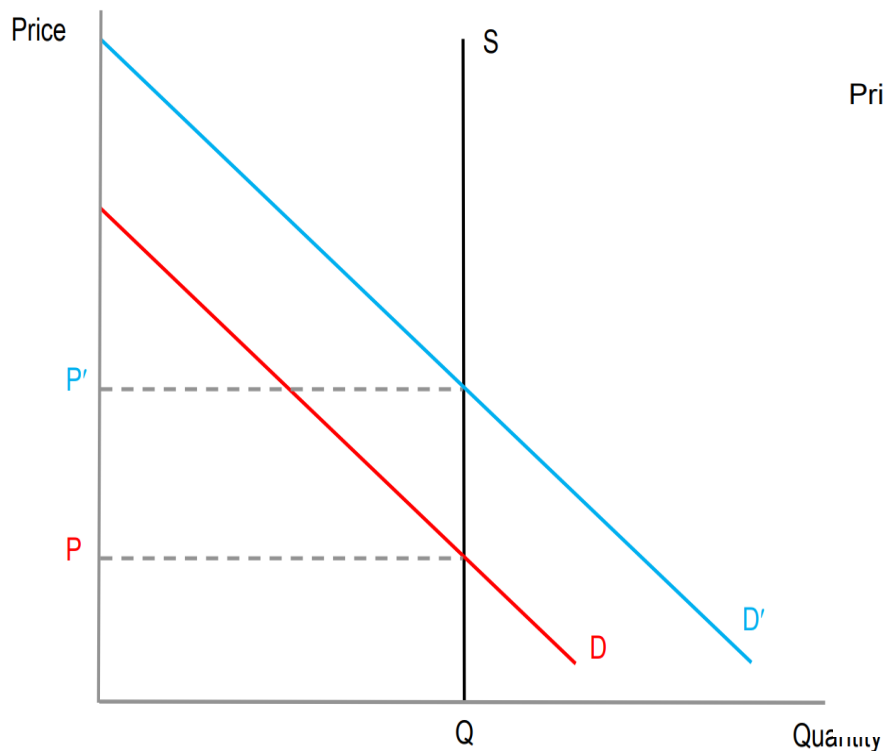
Suppose there are two types of oil wells

- Cheap wells produce oil for \$10/bbl
- Expensive wells produce oil for \$20/bbl

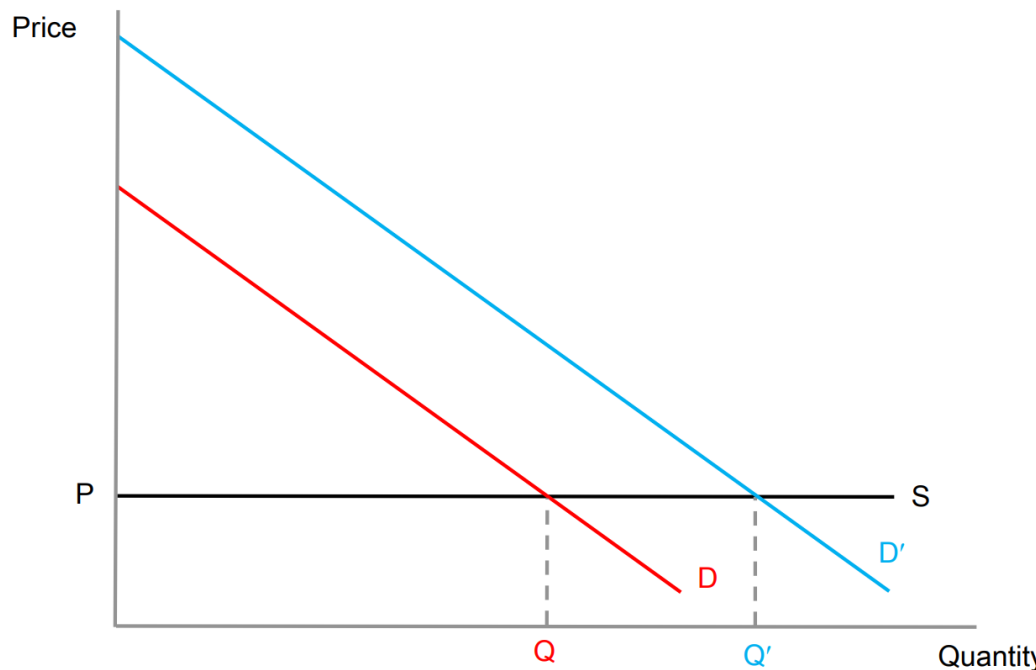


Who captures the rents?

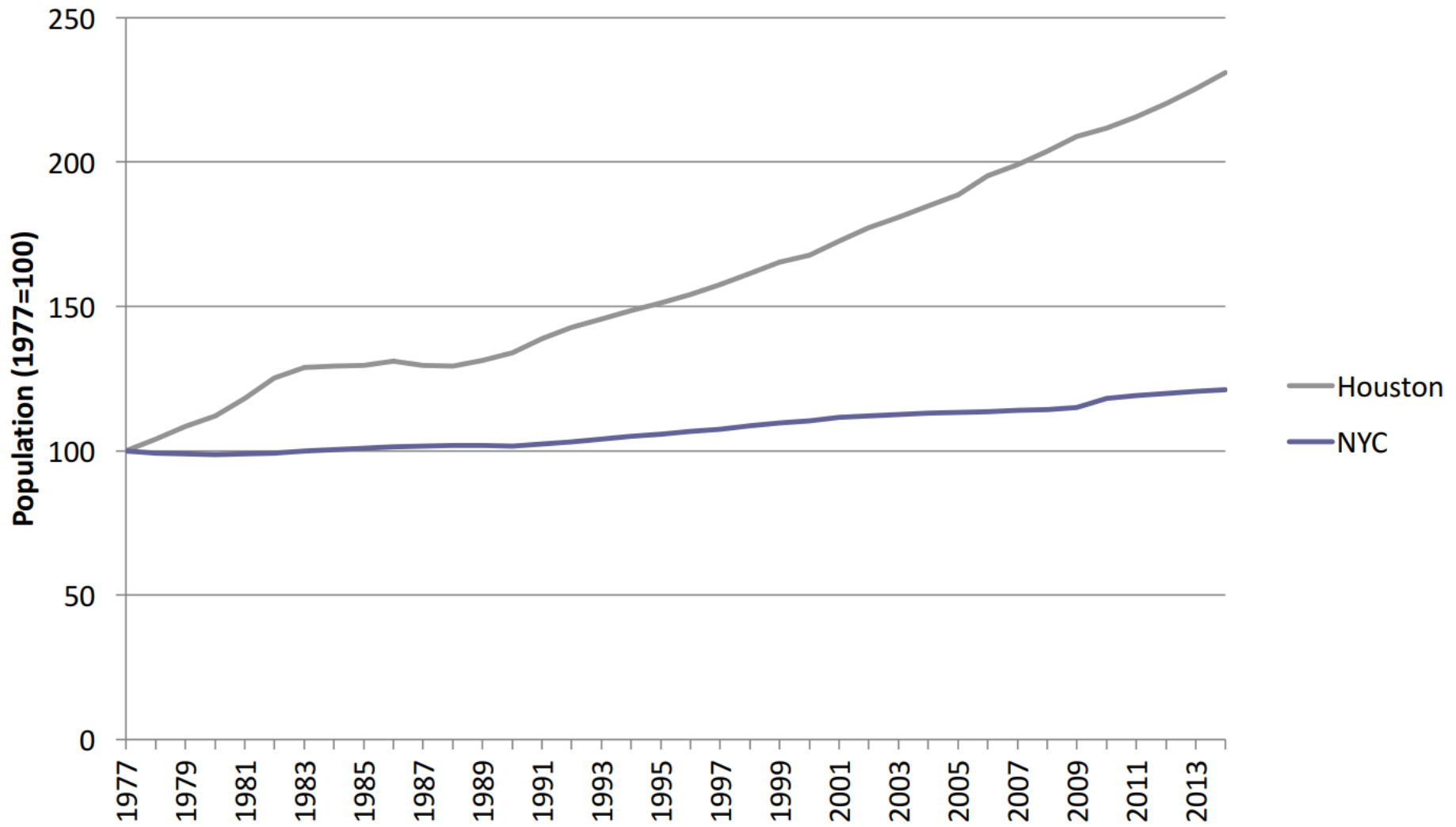
New York



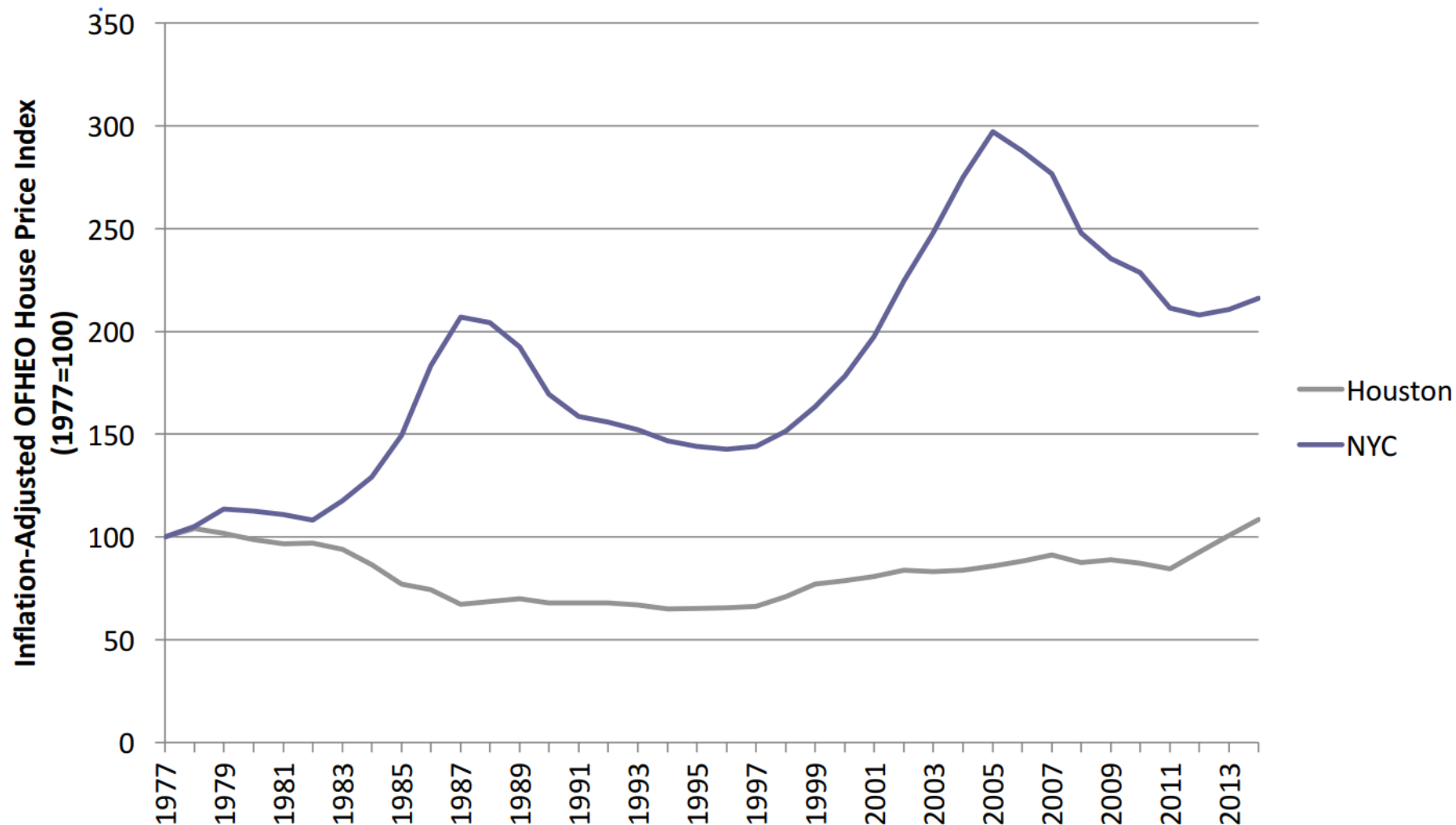
Houston



Population Growth



House Price Growth



Resource-based Advantages: Logic

Sustainability of Resource-based Competitive Advantage requires that:

- Resources are scarce
- Duplication of resources by other firms can be avoided
- Resources are imperfectly mobile

Scarce resources means:

Economic rents go to the owner of the resources and capabilities

Scarce resources

Legal barriers (patents, trademarks, licenses)

Privileged access to inputs, channels, or customers

Caveat: resources must be immobile or cospecialized

This ensures returns go to firm

Counter-example: Professional sports, TV series

“FRIENDS IS LEAVING NETFLIX - AND FANS ARE REELING”

(LA Time headline, Dec 30 2019)



WarnerMedia **paid** \$85 million per year for five years (**\$425 million total**) to land **Friends** for its HBO Max streaming service (outbidding Netflix)

HOME > TV > TV NEWS

'The Office': Why NBCUniversal Is Paying \$500M to Pull the Hit From Netflix

NBC is paying \$500 million to pull 'The Office' from Netflix. Here's why.

BY LESLEY GOLDBERG  JUNE 26, 2019 5:30AM



'The Office' NBC/PHOTOFEST

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Netflix Lands 'Seinfeld' Rights in \$500M-Plus Deal After Losing 'Friends' and 'The Office'

'Seinfeld' is set to be master of a new domain: The iconic comedy will move to Netflix in 2021 as part of a worldwide rights deal.

BY RICK PORTER, LESLEY GOLDBERG SEPTEMBER 16, 2019 9:35AM



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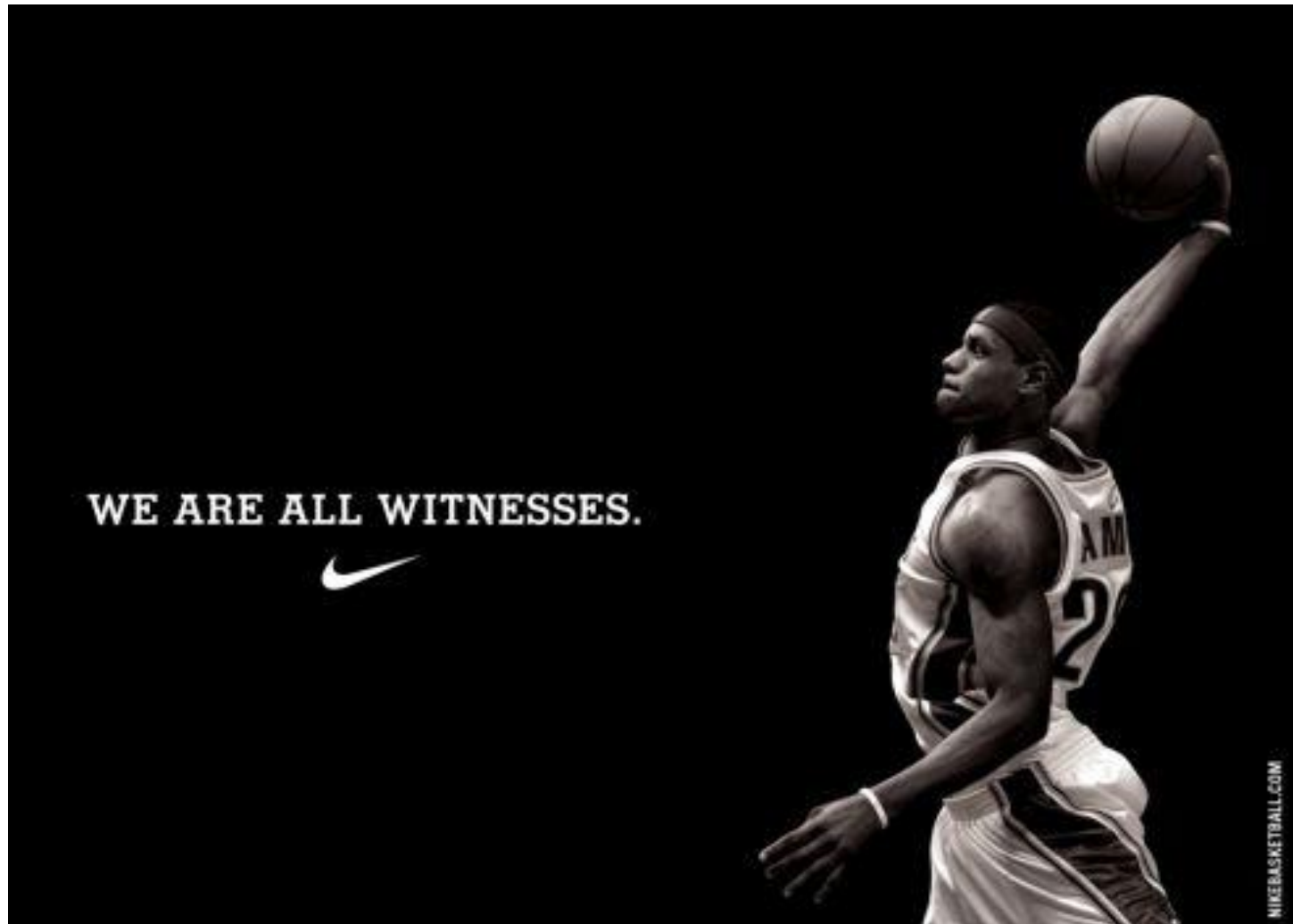
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>\$500M lifetime endorsement contract



Resource-based Advantages: Logic

Mobile resources: Can be traded

- Resources such as Olympic Games, NFL do not generate rents for TV network. [Friends, Office, Seinfeld]

Imperfectly mobile: Co-specialization

One firm has the ability to extract more value from this asset (therefore this firm is WTP more).

- Can be traded, but value inside larger than value outside
 - Nike & LeBron James (>\$500million contract).
 - Disney and Pixar; Disney and Marvel (super heroes)
 - Delta & Atlanta Airport
 - Spotify and Podcasts (Joe Rogan) ???
- Complementarity between Resources

3 Sources of Competitive Advantages

Position-based Competitive Advantage

- Protected by scale and customer captivity
- Often these are “niche strategies” protected by inflexibility of incumbents in adjacent segment.

Capability-based Competitive Advantage

- Learning curve/experience and organizational design.

Resource-based Competitive Advantage

- Resources are scarce, not easily duplicable, and complementary.

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